

Volume II · Chapter 9 — Medical Domain Adaptation · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-9-context.md` for the AI interaction log.

Prerequisites: Ch 7, Ch 8.

Learning outcomes — the student can: distinguish continued pretraining vs. fine-tuning vs. instruction tuning; explain how Meditron/Clinical-Camel/MedGemma were adapted; choose an adaptation strategy; reason about data requirements and licensing.

Topics (theory) — 10 h:

#	Topic	h
9.1	Why adapt general LLMs to medicine; gaps & risks	1
9.2	Continued pretraining on biomedical corpora	2
9.3	Supervised fine-tuning & instruction tuning	2
9.4	Case studies: Meditron, Clinical-Camel, MedGemma, PubMedBERT	2
9.5	Data: sources, curation, quality, licensing for medical corpora	2
9.6	Choosing an adaptation strategy vs. RAG (decision framework)	1

Datasets/tools: HF models (Meditron/MedGemma/PubMedBERT), datasets. **Assessment:** adaptation-plan deliverable (50%); comparison lab (30%); quiz (20%). **Key decisions:** pretrain vs. fine-tune vs. RAG; data licensing; base-model choice. **References:** plan §13 → *Medical LLMs; Fine-tuning, alignment & benchmarks*. **Hours:** Theory 10 + Lab 6 = 16.